

**How do perceptions of Minimum Support Price (MSP) influence
crop choices and risk behaviour among farmers in northern
India?**

Research Paper By Smera Manoj Thakkar

Abstract

Agriculture continues to be the backbone of India's economy, supporting livelihoods, as well as food security for almost half the population and over 1.4 billion people, respectively. The sector, however, continues to be marked by uncertainty, exemplified by market volatility, environmental pressure, and institutional weakness. Given this backdrop, the Minimum Support Price (MSP) architecture, initiated during the Green Revolution era, has turned out to be a cornerstone policy tool to normalize farm revenues as well as assure national food security. Even so, even as MSP is declared for 23 crops, its advantages are skewed towards a handful of staples—largely wheat and rice—and geographically localized in procurement-driven states like Punjab and Haryana. Such skewed implementation has led to high disparities in farmers' perception of MSP, in turn affecting crop selection, risk-taking, and long-term resilience.

This paper analyses the way farmer perceptions of MSP as a definite income guarantee or a distant policy on paper influence farmer decision-making in north India. Case studies of Uttar Pradesh, West Bengal, Punjab, Haryana, and Delhi, the study shows that where procurement is a certainty, farmers take risk-averse strategies and result in monocultures of wheat and paddy. Where procurement infrastructure is poor, farmers resort to distress sales or are induced into taking more risk through diversification. Punjab–Haryana wheat–paddy circuit also reveals the unintended environmental effects of MSP, particularly through groundwater extraction and mass-scale crop residue burning, with externality effects of extreme urban air pollution in Delhi.

The evidence reveals that MSP is not only a price instrument but also a psychological guarantee that influences farmers' behaviour and expectations. Reform of the policies therefore needs to shift from the dramatic price announcements of the past to setting up

credible procurement systems, encouraging diversification, and associating MSP with sustainability outcomes. Finally, farmers' attitudes toward MSP—rather more than the design of policies—dictates whether MSP becomes a farmer livelihood stabilizer or a long-run source of risk and disequilibrium.

1. Introduction

Agriculture has been the economic backbone of India forever, not only shaping the economic trajectory of the country but also the society. Even in present times, the sector is the employer of nearly half of the country's population and central to the food and rural livelihoods of the country. Though the agricultural contribution in the form of GDP has been reducing year after year, it remains the very center of keeping India's over 1.4 billion of the population alive and sustaining the livelihoods of the crores of small and marginal farmers. Nevertheless, despite its value, agriculture in India is marked by an unprecedented level of uncertainty. Market volatility, variability of weather, and increasing input prices together bring in highly volatile incomes, exposing farmers to distress cycles.

To address these challenges, the Government of India introduced the Minimum Support Price (MSP) system in 1966–67 as a part of the Green Revolution reforms. The basic idea behind MSP was to provide farmers with a price floor so that their produce would be bought by state agencies at prices announced by them. This system was meant to protect farmers from sudden drops in market prices and incentivize higher production of staple food grains at a time when India was seriously deficient in them. Over time, MSP became institutionalized as one of the most powerful instruments of agricultural policy, meant to balance consumer demand, farmer well-being, and food security. The system is today extended to 23 crops, comprising cereals, pulses, oilseeds, and commercial crops. Operationally, however, procurement is heavily biased in favour of only two staples—wheat

and rice—both in terms of policy design and historical path dependence (PRS Legislative Research, 2020).

The regional concentration of procurement is also very high. For wheat, three states—Madhya Pradesh, Punjab, and Haryana—grow approximately 46% of the crop but possess 85% of the total procurement. For rice, six states—Punjab, Telangana, Andhra Pradesh, Chhattisgarh, Odisha, and Haryana—grow approximately 40% of total production but possess 74% of procurement. This concentration is indicative of one of the underlying paradoxes of the MSP regime: although MSP is announced on 23 crops in all states, effective gains of procurement are disproportionately accrued by a few states and a small number of crops. Importantly, such concentration goes against the letter and spirit of the National Food Security Act, 2013, which requires governments at the central, state, and local levels to progressively diversify procurement geographically towards the realization of equitable food and nutritional security (PRS Legislative Research, 2020).

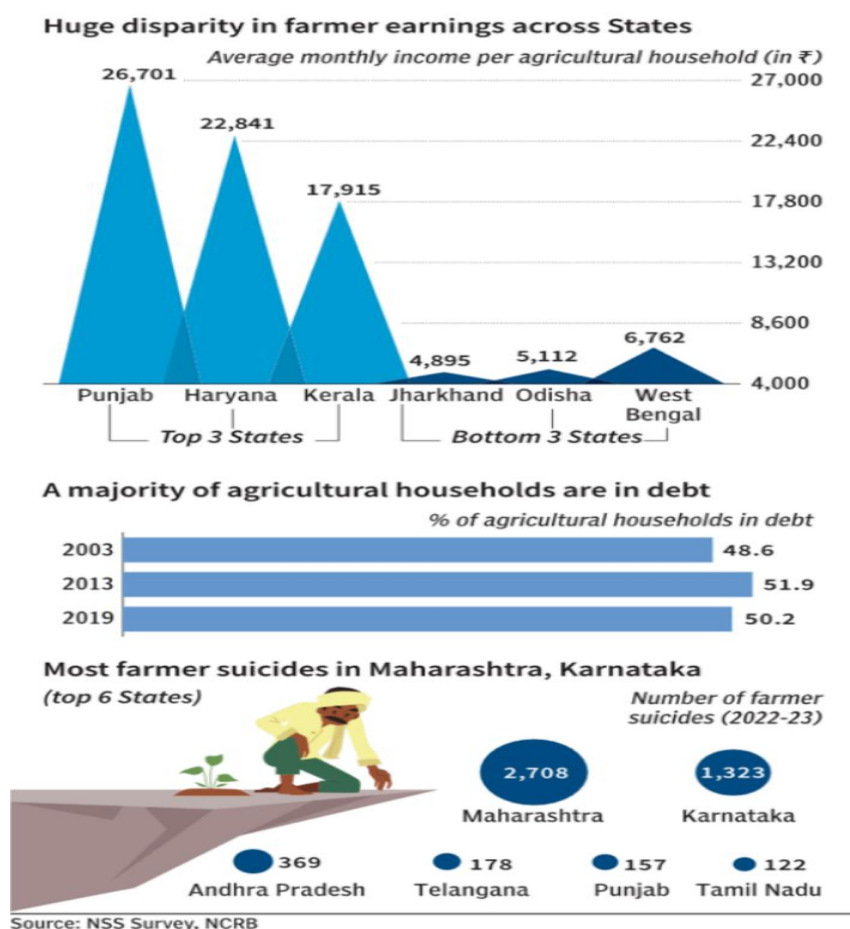
These contrasts indicate that perceptions and implementation, and not price policy alone, constitute MSP. For farmers in states such as Punjab and Haryana, where procurement networks are dense and credible, MSP is perceived as a good assurance of income security. This perception gives rise to a risk-averse attitude, and the consequence is over-reliance on the wheat–paddy cycle in the face of increasing ecological stress from groundwater over-extraction and crop residue burning. In the west and north-east region of India (West Bengal and Bihar), the procurement system is weak and awareness of MSP among farmers is limited. There, the farmers are happy to depend on intermediaries, sell at a price lower than MSP, and face market price fluctuations. Hence, the perception of MSP among them is not of safety but of riskiness, leading to hesitant diversification or continued exposure to market risks (Livemint, 2024). The same effect of perceptions shaping MSP behaviour is observed in the case of laborers.

This deviation implies that the effect of MSP needs to be assessed not only in declared prices but also in terms of credibility, reliability, and accessibility. If MSP is perceived as credible, it achieves stability but at the cost of locking up monocultures. If it is perceived as non-credible, it is a poor risk management tool and farmers will face market price risk exposure. The perception function then mediates between policy intention and ground realities and influences not only crop decisions but also farmer behaviour.

Understanding such dynamics is significant for several reasons. First, it explains the persistence of more ecologically degrading cultivation patterns, e.g., the prevalence of water-guzzling rice in Punjab. Second, it highlights the imperative for policy shift from headline price announcements to prioritize procurement infrastructure, regional balance, and diversification incentives. Third, it highlights the social and political salience of MSP as a tool of not only income support but of farmer confidence, as in mass farmer protests in India in 2020–21 against proposed agricultural market reforms perceived to erode MSP safeguards (The Hindu, 2021).

This paper makes an effort to respond to the central research question: How do farmer attitudes towards MSP influence crop choice and risk-taking among farmers in northern India? Using case study examination of Uttar Pradesh and West Bengal, and Punjab, Haryana, and Delhi, it makes an effort to examine the link between the credibility of MSP and farmer decisions. The analysis substantiates that while MSP is a central policy shaping agricultural realities, ultimately farmer attitudes towards reliability and enforcement decide whether the policy leads to risk-aversion, over-specialization in a few crops, or vulnerability to volatility. The larger implication is that farmer welfare and agricultural sustainability are not simply a matter of MSP rectification but also altering the character of its communication, implementation, and perception.

The graph below shows the wide differentials in farmers' income between states, with Haryana and Punjab way ahead of states like Jharkhand, Odisha, and West Bengal. Over 50% of farm households remain indebted, and farmer suicides—worst in Maharashtra and Karnataka—indicate the severity of agrarian distress. All these indicators point towards the dependence of rural livelihood on institutional safety nets such as MSP.



Source: NSS Survey & NCRB (2023).

2. Background: The Political Economy of MSP

India's most powerful bulwark of agricultural policy in decades has been the Minimum Support Price (MSP), a tool that has been utilized to stabilize rural incomes and attain national food security. To comprehend the impact of its impact on the cropping pattern

and farmer behaviour, it is important to analyse the political economy of the design and implementation of the MSPs. These comprise the history of its construction, institutional processes, geography of trade patterns, and controversies that continue to cloud it.

Historical Roots and Justification

The MSP program was first introduced in 1966–67 at the time of the Green Revolution when India faced severe food shortages and was heavily dependent on imports. In a bid to increase the country's domestic production, the government assured farmers a minimum price for their crops, beginning with wheat and paddy. It had two purposes: it ensured stability in the farmers' income and introduced a procurement facility for the Public Distribution System (PDS), which retailed subsidized food to poor households. Over the years, MSP was extended to 23 crops like pulses, oilseeds, and commercial crops, but in reality, wheat and rice have dominated.

The reasoning behind MSP thus did not exist solely to protect farmers from capricious market forces but also to protect consumers from scarcity. It was never ever an economic instrument alone; it was part of the larger socio-political framework of food self-sufficiency and rural stability.

Institutional Arrangements of MSP

The MSP calculation is a task the Commission for Agricultural Costs and Prices (CACP) performs, advising prices seasonally on the basis of a number of variables. These are cost of cultivation, inter-crop price parity, industry-agriculture terms of trade, and macroeconomic variables. Formally, the formula employed since 2018 is 1.5 times cost of production ($A_2 + FL$), but farmer groups have a tendency to demand a higher degree of comprehensive cost (C_2). The ultimate decision is taken by the Cabinet Committee on Economic Affairs, showing the political nature of the exercise.

In the field, procurement is grossly imbalanced. Over 85% of government procurement is for rice and wheat, and other crops such as pulses and oilseeds are bought in negligible quantities. Even this focus is geographical: three states—Madhya Pradesh, Punjab, and Haryana—grow about 46% of the country's wheat but nearly 85% of procurement; six states also dominate rice procurement although they grow only 40% of the crop in the country (PRS Legislative Research, 2020). This imbalance sustains regional diversity in farmer attitudes towards MSP, with some states considering it a dependable safety net and others considering it a far-off concept.

The Political Economy Dimension

MSP is today a highly politicized institution. Its announcement is generally anchored in the election cycle, with the government appeasing rural electors by enhancing procurement prices. Farmer bodies have always agitated on the issue, most recently during the 2020–21 agitations, when farm reforms proposed were perceived as diluting the MSP guarantee. The scale of such mobilizations, especially in Punjab and Haryana, is testimony to the salience of MSP to farmer morale and livelihood calculations. But political centrality has created issues.

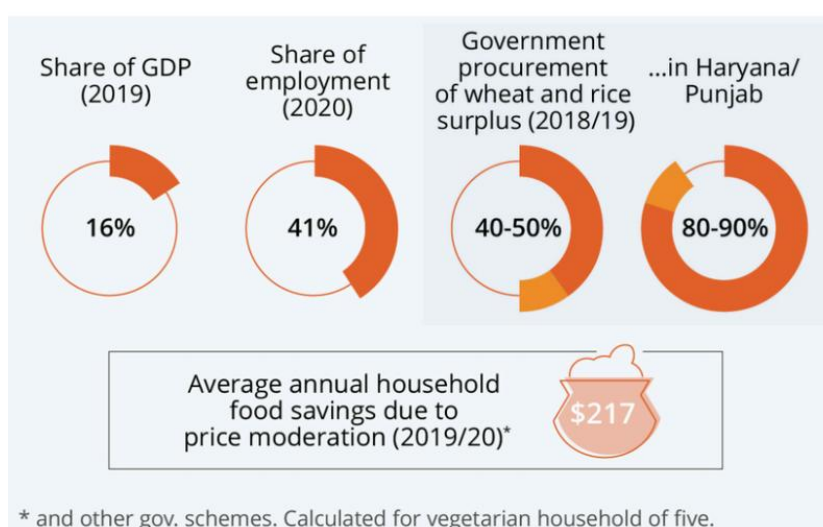
First, there is the cost: procurement, storage, and distribution of food grains impose tremendous pressure on the Food Corporation of India, with inventories often crossing buffer limits.

Second, there are ecological costs: guaranteed procurement of paddy has encouraged monocultures in water-short states like Punjab, where groundwater extraction and burning of crops are now critical concerns (Down to Earth, 2023).

Third, there is the issue of exclusion: while the MSP is very visible, very few of India's 9.3 crore farmers actually sell at these prices, with bigger farmers being the prime beneficiaries.

Small and marginal farmers, who are the bulk, continue to remain dependent on local traders and distress sales.

The significance of MSP cannot be understood in isolation from its role in the larger economy and food security framework. This system helps maintain national food security while supporting rural livelihoods. MSP-backed procurement ensures income security for farmers, especially in surplus-producing states like Punjab and Haryana. The average household food savings of \$217 (₹15,000+) in 2019–20 shows how MSP and related policies help stabilize food prices.



Source: World Bank & *Economic Survey* (2020).

Debates and Alternatives

These have generated controversy for MSP reform. Some farmer organizations demand legal guarantee of MSP for all 23 crops, but the majority of economists warn that this would not be feasible for the exchequer. This guarantee would cost approximately ₹10 lakh crore annually, roughly comparable to the country's overall infrastructure spending (Business Today, 2024). Other schemes such as price deficiency payment schemes—where farmers are compensated if market prices fall below MSP without physical procurement—have been

piloted in states like Madhya Pradesh. These seek to diminish budgetary and storage pressures and yet offer income guarantee but are currently narrow in coverage.

Conclusion

Political economy of MSP reveals the tension between intention and effect. Crafted as a buffer against scarcity and price fluctuation, it has become a security blanket for farmers in procurement-based states and a curse for farmers in non-compliant states. It influences cropping patterns, budgeting decisions, and electoral politics too. Clearly, success is equally dependent on farmers' views on reliability and access and the government policy itself. This background is the framework for analysing why and how these perceptions make a difference to crop selection and risk taking in northern India.

3. Theoretical Framework: Perceptions, Risk, and Behaviour of Farming

Understanding the way farmers determine what to grow and how much risk to take is a system that goes beyond economic price information. Not only in agriculture are behaviours controlled by objective factors like rainfalls, land quality, or market access but also by the subjective perceptions of reliability and trustworthiness and riskiness. The Minimum Support Price (MSP) is a case in point: legally a government-announced minimum price support but in reality depends on the believability and enforceability of the MSP in the minds of farmers.

Risk in Agriculture

Agriculture is risky in the sense that it is exposed to three interrelated types of uncertainty:

Yield risk due to unpredictable weather, insects, or epidemic disease. Price risk, fueled by volatility of demand, supply overhang, and foreign market conditions. Market access risk, due to the presence of intermediaries, transportation challenges, and inefficient procurement channels.

For marginal and small farmers, who make up nearly 86% of India's farmers, these risks are existential. A single crop failure or price meltdown can push households to distress selling and debt. It is here that MSP is supposed to be a stabilizer, ensuring a minimum return irrespective of market fluctuations.

But in practice, the assurance is uncertain. Punjab or Haryana farmers may be very sure that MSP is an assured safety net since procurement is assured and nearly universal for wheat and paddy. But Bihar or West Bengal farmers perceive MSP as uncertain as procurement points are scarce and private traders dominate local markets. The behavioural impact of MSP cannot thus be separated from such local perceptions.

Behavioural Economics and Perceptions

Conventional models of economics assume that farmers behave according to rational reaction to price incentives: higher MSP would mean higher production of the crop. As true as that might be, behavioural economics proves that choices are filtered through institutional credibility and trust perceptions.

For instance:

When the farmers are sure that the state will actually buy at MSP, they are risk-averse and keep on producing wheat and paddy even if diversification can yield them higher returns. Guarantees of procurement are more valuable than the payoff of diversification.

Alternatively, if farmers do not trust MSP implementation, they may shift to high-value crops (pulses, fruits, vegetables) or, more likely, resort to local traders, exposing themselves to fluctuating and typically exploitative prices.

That is characteristic of the kind of path dependence the political economists talk about: getting into the habit of thinking of MSP-supported crops in terms of "safe bets," the

farmers get entrenched in production patterns that are hard to get out of even when environmental or market indicators advise otherwise.

Perception as Risk Management

The perception is germane inasmuch as one views the farmer sentiment with respect to risk. More than a simple economic assessment for the majority of people, risk is emotional and social in addition. Irrigation supported by visible government procurement affords psychological comfort that dampens risk of sustainability. That is why Punjab farmers keep growing water-intensive paddy amidst decreasing groundwater and environmental pressures: the perception of MSP as insurance trumps risk of sustainability (Down to Earth, 2023).

Conversely, ill-managed MSP conditioned farmers typically refer to MSP in terms of "only on paper." Their risk management strategies often involve diversification to various crops, cooperative marketing, or seasonal migration as supplementary incomes. Perceptions of MSP therefore also impact not only crop selection but broader livelihood strategies.

Risk Aversion and Risk Taking

The attitude difference causes the shift in the direction of behaviour:

Risk-Averse Behaviour

Agricultural producers in the well-established procurement regions (Punjab, Haryana, west UP) are extremely risk averse. Such producers choose MSP-covered staples and prefer monoculture. The guarantee of procurement dissuades the producers from diversification and experimenting with other crops like maize, pulses, or oilseeds even when the latter are well-suited to regional agro-climatic parameters.

This results in revenue stability but introduces long-run economic and environmental imbalances such as stubble burning and excessive stocking of goods in government warehouses.

Risk-Taking or Forced Exposure

Farmers in weaker procurement networks (West Bengal, Bihar) face greater volatility. They diversify into niche crops or horticulture, which have higher potential prices but risk. Others continue to rely on intermediaries, in practice being forced to take risks with no institutional safety net.

MSP perceptions of unreliability here not only drive economic insecurity but also increase farmer distress.

The Social Network and Collective Action Role

Perceptions are not built in isolation; they are built by webs of local power institutions, farmer unions, and communities. Punjab and Haryana's strong farmer unions form the basis of the perception of MSP as a right to be asserted, as seen in the 2020-21 agitations. Where collective organization is weaker in states, MSP is less seen as a guarantee and more as a symbolic assertion. This difference highlights how institutional presence and social mobilization frame the credibility of MSP among farmers (PRS Legislative Research, 2020).

Why Perceptions Matter

The worth of perceptions is that they bridge policy intention and farmer reality. Announcing MSP for 23 crops is not automatically going to change farmer behaviour until farmers believe procurement is credible and within reach. Perceptions therefore act as a filter, and convert written policy into experience that is felt. Perceptions account for why two states

with identical MSPs for wheat—Uttar Pradesh and West Bengal—exhibit such different patterns of price stability and farmer confidence.

Conclusion

The theoretical model of capturing MSP's impact on farmer decisions will therefore need to reconcile economics, psychology, and politics. It must also presume that the farmers are not just margin price takers but are agents embedded in institutional environments that shape risk and reward perceptions. Where MSP is taken as exogenously given, farmers take risk-averse decisions on the basis of wheat and paddy; where taken to be uncertain, they are either coerced into diversification or left vulnerable to perilous markets. Perception of MSP is therefore a voiceless umpire of crop decisions and risk taking and is therefore the focal point of any consequential analysis of agricultural policy across northern India.

4. Case Study I: Wheat Markets of Uttar Pradesh and West Bengal

The case of UP vis-à-vis West Bengal is in striking contrast with the existence of statewide variation in the attitude of MSP and the salience of attitudes in the risk behaviour and crop choice of the farmers.

We have two major wheat-producing states, but the institutional environment, procurement regime, and farmer experience with MSP vary substantially. This variation illustrates that the same national policy will have very different consequences based on the local interpretation and implementation of it.

Although West Bengal is geographically outside northern India, it provides a useful contrast to Uttar Pradesh. Both states produce wheat, yet their procurement structures and farmer experiences with MSP are very different. Uttar Pradesh represents a case where MSP is perceived as credible due to stronger mandi regulation and government procurement, while West Bengal illustrates how weak procurement infrastructure erodes trust in MSP. Including

this comparison highlights that farmer behaviour depends less on the MSP announcement itself and more on perceptions shaped by state-level implementation. This contrast sharpens the analysis of how perceptions influence crop choices and risk-taking behaviour.

Agricultural Background of Uttar Pradesh and West Bengal

Uttar Pradesh is India's largest wheat-producing state, with more than 30 million tonnes annually produced. It boasts centuries-old wheat production based on extensive irrigation systems in the west, popularly known as the "wheat bowl of India." West Bengal, being one of the largest agricultural states, also has a more diverse crop basket, where rice is the leading crop in more than one season (Aus, Aman, and Boro). Wheat farming in West Bengal is much smaller in size, confined to a few districts, and has historically been dwarfed by paddy and jute.

These structural differences provide the basis upon which MSP is implemented in each state. In UP, the leadership of wheat is based on government procurement priorities, while in West Bengal the stress on rice and the smaller land holdings dilute the impact of wheat MSP.

Institutional Structures: APMCs and Intermediaries

The difference begins with institutional arrangement in marketing farm products. Uttar Pradesh possesses a relatively stronger infrastructure of Agricultural Produce and Marketing Committees (APMCs), that regulate mandis (local agri-markets). Procurement in mandis is direct and transparent by government agencies. Farmers sell wheat to regulated mandis, where procurement units of the Food Corporation of India (FCI) or state agencies purchase grain at MSP. Government buyer participation puts pressure on prices since private traders are forced to quote at least comparable prices. This strengthens farmer trust in MSP, which becomes a more credible "real" price floor than a symbolic policy.

West Bengal, on the other hand, has comparatively less restrictive APMC regulation. Procurement is limited and middlemen-dominated mandis buy directly from farmers. As government procurement centers are few, farmers have to sell to private traders at rates that are frequently below MSP. Hence, farmers view MSP as unreliable. MSP is on paper but rarely turns into real income protection for them.

The institutional strength of APMCs thus dictates how perceptions are constructed. While farmers view visible procurement to internalize MSP as actual, without procurement they deny it as speculative.

Price Volatility and Stability

The week-on-week series of wheat prices for these two states reflects the impact of MSP on volatility.

Wheat prices in Uttar Pradesh do not fluctuate significantly from one week to another. This is a result of the stabilizing influence of MSP-supported procurement in mandies. Even if private traders corner part of the local markets, the presence of procurement centers prevents them from pushing prices too far below MSP since farmers can sell to government buyers otherwise.

In West Bengal, in contrast, price fluctuations are more abrupt. Lacking guaranteed procurement, prices respond promptly to seasonal supply peaks and dealer initiatives. Farmers are often compelled towards distress sales during harvest periods, to be followed by off-season peak prices. Volatility destroys income anticipatability and contributes to farmer distrust of the system.

This heterogeneity validates that MSP's effect is not the same everywhere; it varies based on whether farmers believe that procurement will indeed protect them from price shocks.

Risk Behaviour and Crop Selection

MSP perceptions are translated into farmer behaviour toward risk taking and crop selection directly.

Uttar Pradesh (risk-averse behaviour): UP farmers, particularly in western regions, are risk-averse. With MSP-based procurement guaranteed to them, they continue to sow wheat when diversification into pulses or oilseeds would be profitable. Wheat is a "sure bet" government-supported for them. It is for this reason that we have the long-standing wheat monoculture in the region, which is climatically desirable as well as policy-induced risk aversion.

West Bengal (higher risk exposure)

West Bengal farmers are more uncertain. With weak procurement networks, they cannot rely on MSP to stabilize prices. Their options then reflect more market risk exposure. Some farmers diversify into vegetables, fruits, or specialty crops in hopes of getting better prices, while others are stuck with intermediaries and sell their wheat or rice at prices below MSP. In either case, suspicion of MSP forces them to take more risk than their UP counterparts.

Effect on Income Security

The gap in perceptions has far-reaching consequences for farm revenues. In UP, the availability of procurement centers ensures that a majority of marketed wheat is sold at or near MSP, bringing assured support to revenues. In West Bengal, farmers will be more likely to be quoted prices much lower than MSP through distress sales. This not only lowers mean incomes, but also increases income volatility, making household financial planning more susceptible.

Most importantly, West Bengal's average holdings are smaller. This makes the problem worse. With fewer surpluses available to sell, farmers lack bargaining power, and their ability to carry stocks until prices recover is limited. The structural and institutional handicaps thus accumulate, making MSP an ineffective safety net.

Political and Social Implications

These alternative perceptions also get translated into political mobilization. MSP is a political issue, and farmer organizations tend to bargain with the state for remunerative prices and timely procurement in UP, especially western UP. MSP is less of a concern in West Bengal because of partly low expectations. This differential mobilization also adds to the divide: strong farmer movements reinforce the legitimacy of MSP somewhere and its absence reinforces weak perceptions somewhere else.

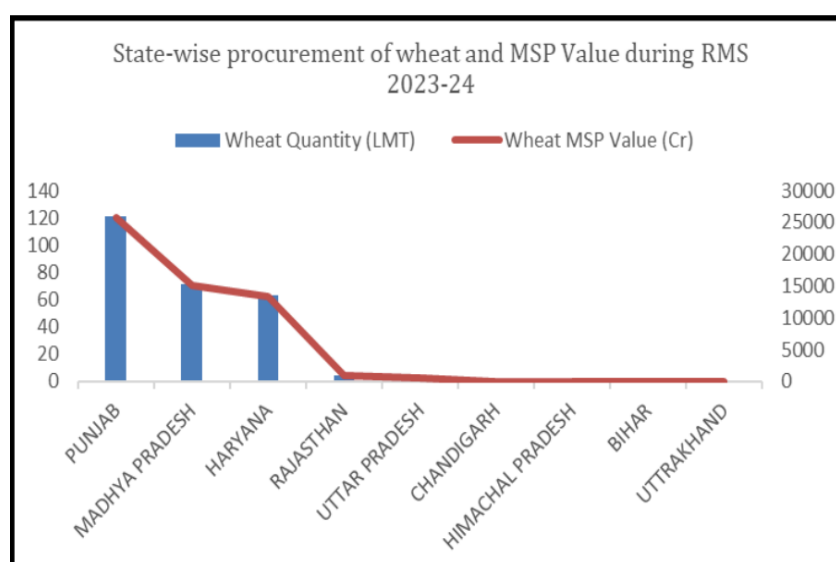
Conclusion: Lessons from the Comparison

The UP–West Bengal contrast reveals a hidden reality: the success of MSP is not only a question of national declarations but also one of institutional ability at the state level, and of farmers' perceptions of credibility. Transparent procurement in UP generates trust, leading to risk-averse crop choice around wheat. Inefficient procurement in West Bengal generates distrust, subjecting farmers to volatility and driving them towards riskier or distress-based alternatives. The moral of the story is that national policy consistency does not always mean consistency of outcome. Unless procurement facilities are enhanced and regional disparity removed, MSP continues to be a security symbol for some farmers and a hollow promise to others.

5. Case Study II: Punjab, Haryana, and Delhi – MSP and Crop Residue Burning

Punjab and Haryana have been India's largest beneficiaries of its MSP policy since its launch. Both contribute over one-third of the wheat and rice to the pool. But with this success story comes a shadow: the huge wheat and paddy production volume has resulted in ecologically unsustainable trends, especially mass crop residue burning. This case study describes how the MSP influences cropping patterns in Punjab and Haryana, how the patterns influence crop residue burning (CRB), and how the ecological cost is borne by Delhi and elsewhere.

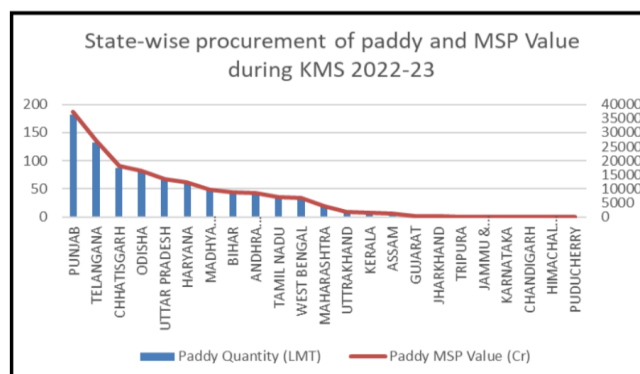
State-wise Procurement of Wheat and MSP Value, 2023–24



Source: Ministry of Consumer Affairs, Food & Public Distribution (2023).

Punjab, Madhya Pradesh, and Haryana together account for the overwhelming majority of wheat procurement, with Uttar Pradesh contributing only a modest share and states like Bihar almost absent. Such concentration reinforces regional imbalances in how MSP is perceived and utilized.

State-wise Procurement of Paddy and MSP Value, 2022–23



Source: Ministry of Consumer Affairs, Food & Public Distribution (2023).

Punjab, Telangana, and Chhattisgarh dominate paddy procurement, while states like Bihar and Jharkhand contribute minimally despite large farmer populations. This reflects the unequal reach of MSP-backed procurement, which leaves many eastern farmers dependent on private intermediaries.

MSP and the Paddy–Wheat Cycle

The origins of the cropping pattern of Punjab and Haryana date back to the Green Revolution of the late 1960s. Lured by irrigation facilities, high-yielding seed hybrids, and above all, guaranteed procurement under MSP, farmers moved unequivocally towards a paddy–wheat cycle. Entirely more than 85% of Punjab's cultivable area is today occupied by these two crops, though it is known that they are not ecologically adapted to the semi-arid climate of the region.

It is MSP that is the main reason for this monoculture. Wheat and paddy are "safe" crops for farmers since almost all surplus marketed is bought at MSP from a series of closely spread procurement centers and mandis. Such predictability deters diversification into pulses, oilseeds, or horticulture even when such crops may be more appropriate to soil and water

conditions. The notion of MSP as a certain safety net thus induces risk-averse behaviour, and the paddy–wheat cycle goes on and on.

The Issue of Crop Residue

Paddy crops produce massive quantities of crop residue in the shape of straw. Punjab alone produces over 20 million tonnes of paddy residue annually. Farmers have just 15–20 days between harvesting of the paddy crop and sowing of wheat. Due to the lack of labor and the costly nature of residue management equipment, field burning of crop residue is the most common practice.

Burning of residue has disastrous environmental consequences. It releases high amounts of particulate matter (PM_{2.5}), carbon monoxide, and greenhouse gases. It contributes largely to Delhi's infamous smog and the National Capital Region (NCR) every winter. Crop burning adds about 30–40% of Delhi's PM_{2.5} load during peak burning seasons (Down to Earth, 2023). Public health consequences are enormous, with rising cases of respiratory diseases and falling air quality indexes.

MSP Role in Residue Burning

The link between MSP and residue burning is in its incentive system:

- A high MSP of paddy encourages overproduction of a crop that generates enormous quantities of residue.
- Purchasing commitments assure farmers that they have minimal financial risk in repeating this, no matter how detrimental it is to the environment.
- Wheat's narrow sowing window, driven by MSP, does not leave any room for natural decay or environmentally friendly disposal of residue.

In reality, the same system designed to reduce economic risk ends up increasing ecological risk. The farmer does not neglect the cost to the environment, but in the absence of the inexpensive alternatives, he chooses the inexpensive, short-term option of burning.

Impact on Farmer Behaviour

MSP perceptions influence farmer decisions in two significant ways:

a. Risk Aversion in Crop Choice

Haryana and Punjab farmers barely diversify to other crops even in the event of water shortages and land degradation. Maize, for example, is less water-intensive and would reduce crop residue burning, but is without the backing of the MSP guarantee. Without MSP guarantee, farmers consider maize risky, although it is agronomically good.

b. Risk-Taking in Ecological Practices

Ironically, the same risk-averse crop-chosen farmers become risk-takers in environmental practice. Residue burning is technically illegal and sparks public ire, but farmers persist in doing so because the economic cost of non-compliance is low and the economic cost of alternatives is high. The perceived MSP-supported income security allows them to afford the risk of fines or social retribution.

Policy Responses and Constraints

The government has made various attempts in residue burning control. Subsidy to technology such as the Happy Seeder, in which sowing is made possible without the need for burning residue, has been provided. Fines and penalties for burning have been imposed. Adoption is still low because such alternatives are seen as expensive, labor-intensive, or not viable due to lack of time.

A few reports suggest that higher MSP or bonus on crops like maize or pulses can induce diversification and thereby indirectly deter burning of residues (PRS Legislative Research, 2020). These reforms have been politically difficult, however, because farmer unions resist any reform that is perceived to water down wheat–paddy procurement. The perception problem thus persists: farmers won't change their ways unless they believe that paddy alternatives will be just as secure as far as procurement and profit are concerned.

The Delhi Connection

Delhi, being a neighboring city, is the frontline sufferer of Punjab and Haryana's farming. Every year, October and November witness the city's air quality hit rock bottom as crop residue burning smoke blends with city pollution. It has turned MSP into a national public health issue rather than a rural income policy. The issue is an example of how farmer attitudes towards MSP, influenced by local procurement conditions, extend far from the farm gate.

Conclusion: Risk Transfer from Farmers to Society

The Punjab–Haryana case demonstrates the manner in which MSP affects risk in a multi-faceted way. MSP reduces income risk and stabilizes farmers. However, it encourages risk-averse crop choice (wheat and paddy overdependence) and, in combination, encourages risky environmental practice (residue burning). The social opportunity cost of the risk transfer is incurred by society in general, and Delhi and the NCR suffer falling air quality and negative health effects. The warning is loud and clear: unless MSP is transformed to aid diversification and provide genuine choice, attitudes towards MSP as a safety net will reinforce harmful ecological practices. The answer is in a joined-up approach that unites income support with ecological sustainability and ties farmer incentives with broader social outcomes.

6. Synthesis of Findings

The case studies of MSP and the impact of the MSP on the behaviour of the farmers of north India exhibit a multifaceted mix of economic incentives, institutional parameters, and farmer attitudes.

Perceptions as Selective Filters

One of the main results of this study is that MSP effect depends more on whether farmers believe procurement is going to happen and less on the announcement of the news in the headlines. Government procurement that is proven in Uttar Pradesh builds up confidence and risk-avoiding attitude by farmers with respect to wheat. Poor procurement breaks the confidence and brings about dependence on intermediaries and exposure to volatility by farmers in West Bengal. MSP has created a very entrenched wheat–paddy cycle in Punjab and Haryana, but in Delhi the effect is through air pollution.

That is, MSP is more than a economic safety net but a psychological insurance tool that has a bearing on the expectations and choice of farmers.

Divergent Risk Behaviour

The MSP's point of view evokes two opposite behaviours:

- Risk-averse procurement-strong state strategies (Punjab, Haryana, western UP), where farmers prefer to hold on to MSP-backed staples at environmental cost.
- Risk-hit or forced diversification in weaker procurement states (Bihar, West Bengal), where farmers attempt alternative crops or distress sales at prices below MSP.

This divergence reflects the way uneven implementation of MSP generates two agricultures in India: one stabilized with the support of the state and the other exposed to market vagaries.

Ecological Trade-offs

Punjab and Haryana demonstrate how MSP reduces earning risk but causes environmental risk. MSP's push for paddy leads to groundwater depletion, soil stress, and massive crop residue burning. MSP largely contributes to Delhi winter smog, and it is farmers' risk transfer to society (Down to Earth, 2023).

Regional Imbalances

Procurement is unevenly distributed: three states buy 85% of wheat and six states buy nearly three-quarters of rice (PRS Legislative Research, 2020). This generates huge differences in the farmers' perception of MSP across India. MSP is taken for granted in Punjab, whereas in West Bengal it is seen as far away and not trustworthy.

7. Policy Implications

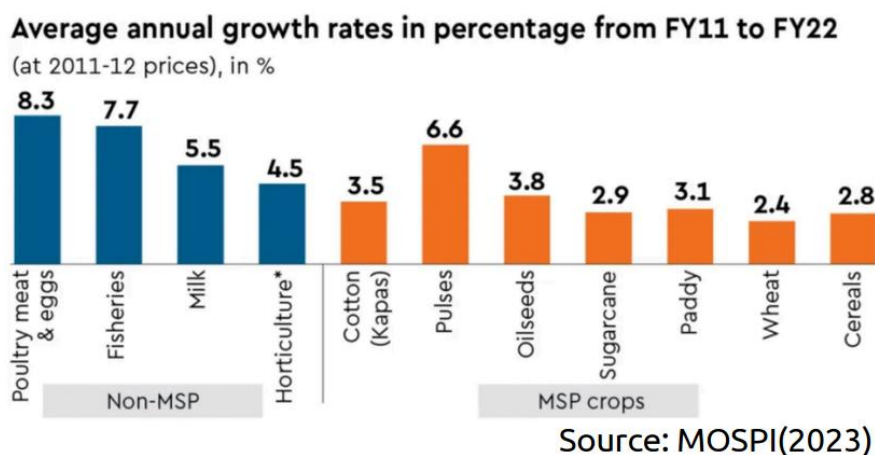
The report says that MSP cannot be singly concentrated on procurement of wheat and rice in particular states. Reform has to monitor three goals simultaneously: farmer well-being, environmental sustainability, and regional equilibrium.

Diversify Beyond Wheat and Paddy

Procurement needs to be increased substantially to pulses, oilseeds, and millets. Diversification would conserve water, improve soil health, and be more consistent with Indian nutritional needs. Farmers will diversify only if procurement systems are perceived to be benefiting these crops.

The following figure illustrates that non-MSP commodities such as poultry, fisheries, and dairy have recorded much faster growth in price than MSP-backed cereals. In contrast, wheat and paddy — heavily supported through procurement — show relatively low growth. This divergence highlights how the MSP system can lead to food inflation. Price hikes in

commodities like soybean and maize, which are basic feed material for poultry, fishery and dairy, will push the prices of such products, leading to high food inflation (Financial Express, 2024).



Source: MOSPI (2023).

Enhance State-Level Procurement Systems

West Bengal and Bihar require more regulated mandis and procurement centers. Institutional presence is needed otherwise MSP is "on paper." Decentralized procurement, emulating robust states, can construct credibility where there is none.

Align MSP with Sustainability Objectives

Reforms can include environmental incentives. Bonus payments, for example, can be made to reward farmers who grow water-low crops or whose farms have no residues. Linking purchases to sustainability would reduce the environmental risks created by current trends.

Investigate Alternative Models of Income Support

Price Deficiency Payment (PDP): Payment to farmers whenever market prices fall below MSP without physical procurement. Fiscal and storage costs are avoided with income security assured. Madhya Pradesh's Bhavantar Bhugtan Yojana is a case in point.

Direct Support of Income

Schemes such as PM-KISAN involve assured payments. While effective as they are, they do not give the psychological guarantee of a floor price. A package of MSP purchase, PDP, and income transfer can provide a middle ground.

Engage Small-scale Farmers More Equitably

Now, only 6–9% of Indian farmers get MSP prices (Shanta Kumar Committee, 2015). Most of the benefits accrue to large farmers (Shanta Kumar Committee, 2015). Developments such as doorstep procurement, aggregation through FPOs, and digital payment mechanisms can reduce barriers to smallholders, making MSP more inclusive.

Close the Information Gap

Perceptions of MSP are as much knowledge-based as reality-based. In weak procurement states, most farmers don't know procedures or doubt payments on time. Greater communication and transparency would make the system more credible.

8. Broader Impacts to Food Security and Society

From Food Security to Sustainability

MSP has assured India's grain buffer, but this success is more and more vulnerable. FCI godowns are filled to the brim with excess rice and wheat, while groundwater is being depleted and environmental costs are accumulating. Food security needs to transform to nutrition and sustainability security, on diversified, climate-resilient crops instead of on narrow monocultures.

Political Sensitivity of Reform

Any MSP reform will have to acknowledge its political consequences. Farmers don't just regard MSP as a policy but a right. The agitations of 2020–21 showed that the kind of reforms viewed as undermining MSP can result in mass mobilization. Reform in this situation thus needs incremental, inclusive measures that offer concrete alternatives prior to altering procurement priority.

Urban-Rural Interdependence

The Delhi case shows how rural policy determined by MSP affects urban public health. Crop residue burning is one of the proximate reasons for the capital's winter air pollution crisis. It shows that MSP is not only a rural or agricultural policy but a national social and environmental policy.

9. Path Forward: Towards an Integrated MSP

In order to reconcile farmer well-being with environmental and cost considerations, MSP has to become a complete system:

Economic integration: Align MSP with income transfer and crop insurance to get multi-layered protection.

Ecological integration: Associate procurement with sustainable environmental practices such as crop rotation, residue management, and water management.

Regional integration: Boost procurement from other regions of eastern and central India in order to diversify geographical concentration.

Institutional linkage: Link cooperatives and FPOs in procurement and reduce dependence on the big landholders and ensure smallholder inclusion.

Such a consolidated MSP would transcend its limited function of a price support tool and turn into a true instrument of agricultural future making.

10. Conclusion

This paper has studied the impact of awareness of MSP on crop selection and farmer riskiness in north India. The study results suggest that the impact of MSP is mediated, not only by prices but also by procurement credibility and trust. In Uttar Pradesh, it causes risk-averse wheat monoculture with strong procurement. In West Bengal, weak procurement breaks the trust and brings volatility to the farmer. In Punjab and Haryana, the MSP stabilizes the farmer incomes but causes the wheat–paddy cycle with the production of ecological spillovers such as residue burning that chokes the city of Delhi's atmosphere.

MSP hence covers against earnings risk but introduces fresh risks in the form of regional disequilibrium, environmental degradation, and fiscal strain. It is a system paradoxically designed to protect farmers that has the promise of threatening long-term sustainability unless it is reinterpreted. The future does not eliminate MSP but interprets it differently. Diversification, decentralisation, environmental incentives, and smallholder inclusion can make MSP more sustainable and more inclusive.

Interestingly, reform is also required to accept that perception is reality itself. Until farmers feel that procurement is fair and reasonable, the largest announcements of MSP will be irrelevant in terms of making a difference in the behaviour of farmers. MSP is more of a social contract between the state and the farmer and less of a mathematical system of prices in the main. It has to be redefined not only for the purpose of farmer welfare but also for the larger purposes of equity, sustainability, and resilience of the agricultural economy in India.

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